

```

> restart: unprotect('D'):

m := 5;

twist := - 7;

d := 13;                      degree := d + 1:

##

#SS := x^(d+1) + 1^(d+1) + 0*x^6:
#HSS := X^(d+1) + T^(d+1) + 0*X^6:

SS := x^(d+1) + 1^(d+1) + 1*x^6:
HSS := X^(d+1) + T^(d+1) + 1*T^(d+1-6)*X^6:

##

Jet_xR := expand(
+ (1/Ry^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (R1/RR)^j *
Delta_xR^0, j=0..m-0*3)
+ (1/Ry^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (R1/RR)^j *
Delta_xR^1, j=0..m-1*3)
+ (1/Ry^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (R1/RR)^j *
Delta_xR^2, j=0..m-2*3)
+ (1/Ry^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (R1/RR)^j *
Delta_xR^3, j=0..m-3*3)
+ (1/Ry^(m-4*2)) * add(EE||j * x1^(m-4*3-j) * (R1/RR)^j *
Delta_xR^4, j=0..m-4*3) ):

##

for l from 0 to m - 0*3 do dA[l] := (m-0*2)*d - m + l*1 + 0*1 +
twist: od:

for l from 0 to m - 1*3 do dB[l] := (m-1*2)*d - m + l*1 + 1*1 +
twist: od:

for l from 0 to m - 2*3 do dC[l] := (m-2*2)*d - m + l*1 + 2*1 +
twist: od:

for l from 0 to m - 3*3 do dD[l] := (m-3*2)*d - m + l*1 + 3*1 +
twist: od:

for l from 0 to m - 4*3 do dE[l] := (m-4*2)*d - m + l*1 + 4*1 +
twist: od:

##

```

```

Delta_xR := x1 * (R2/RR - R1*R1/RR^2) - x2 * (R1/RR):
Solx1 := - y1*Ry/Rx + R1/Rx:
Solx2 := - Rxx*y1^2*Ry^2/Rx^3 + 2*Rxx*y1*Ry*R1/Rx^3 - Rxx*
R1^2/Rx^3 + 2*y1^2*Rxy*Ry/Rx^2
          - 2*y1*Rxy*R1/Rx^2 - y1^2*Ryy/Rx - y2*Ry/Rx + R2/Rx:
##
Jet_Ry := sort(mtaylor(
          expand(subs(R2=-(RR^2*Delta_Ry-RR*R1*y2-
R1^2*y1)/(RR*y1),
          expand(subs({x1=Solx1,x2=Solx2},
Jet_xR))),
          [R1,y1,Delta_Ry], 100), [R1,y1,Delta_Ry]):
##
printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do
EEq[m-j-3*k,j,k] := factor(
  Ry^(m-j-3*k)*RR^(m-j-3*k)*Rx^(m-2*k)
*
  coeftayl(Jet_Ry, [R1,y1,Delta_Ry]=[0,0,0], [m-j-3*k,j,k]) ):
od: od:
## EQUATION RR EXPLICITE
RR := y^(d+1) + SS;
HRR := Y^(d+1) + HSS;
RY := diff(HRR, Y):
Rx := diff(RR,x):
Ry := diff(RR,y):
Rxx := diff(RR,x,x):
Rxy := diff(RR,x,y):
Ryy := diff(RR,y,y):
##
printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do
Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], y, (m-j-3*k)*d):
Var[m-j-3*k,j,k] := indets(Eq[m-j-3*k,j,k]) minus {x,y}:

```

```
od: od:
```

```
##
```

```
m := 5
```

```
twist := -7
```

```
d := 13
```

```
RR :=  $x^{14} + y^{14} + x^6 + 1$ 
```

```
HRR :=  $T^{14} + T^8 X^6 + X^{14} + Y^{14}$ 
```

(1)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(1*d)}$ 
```

```
for l from 0 to m - 0*3 do  
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA  
[l],1*d-1)): od:
```

```
for l from 0 to m - 1*3 do  
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB  
[l],1*d-1)): od:
```

```
for l from 0 to m - 2*3 do  
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC  
[l],1*d-1)): od:
```

```
for l from 0 to m - 3*3 do  
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD  
[l],1*d-1)): od:
```

```
for l from 0 to m - 4*3 do  
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE  
[l],1*d-1)): od:
```

```
## SYSTEM FOR DIVISIBILITY BY  $y^{(1*d)}$ 
```

```
Systemeyld := {}:
```

```
for k from 0 to (m-1)/3 do
```

```
Coeffs_yld[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,  
k], y, 1*d)), [x,y])}:  
Systemeyld := Systemeyld union Coeffs_yld[1,m-1-3*k,k]:
```

```
od:
```

```
Systeme_yld := Systemeyld:
```

```
## VARIABLES FOR DIVISIBILITY BY  $y^{(1*d)}$ 
```

```
Variables_yld := indets(Systeme_yld):
```

```

## RESOLUTION  $y^{(1*d)}$ 

Solutions_y1d := solve(Systeme_y1d, Variables_y1d):

assign(Solutions_y1d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

$l := 0$

2888

$l := 1$

5282

$l := 2$

650

$l := 3$

663

$l := 4$

676

$l := 5$

689

$l := 6$

$l := 0$

368

$l := 1$

512

$l := 2$

325

$l := 3$

$l := 0$

$l := 0$

$l := 0$

(2)

> ## EXPAND ALL UNKNOWNNS UP TO $y^{(2*d)}$

```

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],2*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],2*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],2*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],2*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],2*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(2*d)

Systemey2d := {}:

for k from 0 to (m-2)/3 do

Coeffs_y2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
k], y, 2*d)), [x,y])}:

Systemey2d := Systemey2d union Coeffs_y2d[2,m-2-3*k,k]:

od:

Systeme_y2d := Systemey2d:

## VARIABLES FOR DIVISIBILITY BY y^(2*d)

Variables_y2d := indets(Systeme_y2d):

## RESOLUTION y^(2*d)

Solutions_y2d := solve(Systeme_y2d, Variables_y2d):

assign(Solutions_y2d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

l := 0

```

l := 1
5275
l := 2
9772
l := 3
1157
l := 4
1183
l := 5
1209
l := 6
l := 0
30
l := 1
62
l := 2
84
l := 3
l := 0
l := 0
l := 0

```

(3)

```

> ## EXPAND ALL UNKNOWNNS UP TO y^(3*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],3*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],3*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],3*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],3*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],3*d-1)): od:

```

```

## SYSTEM FOR DIVISIBILITY BY  $y^{(3*d)}$ 

Systemey3d := {}:
for k from 0 to (m-3)/3 do
  Coeffs_y3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], y, 3*d)), [x,y])}:
Systemey3d := Systemey3d union Coeffs_y3d[3,m-3-3*k,k]:
od:
Systeme_y3d := Systemey3d:
## VARIABLES FOR DIVISIBILITY BY  $y^{(3*d)}$ 
Variables_y3d := indets(Systeme_y3d):
## RESOLUTION  $y^{(3*d)}$ 
Solutions_y3d := solve(Systeme_y3d, Variables_y3d):
assign(Solutions_y3d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

$l := 0$

3538

$l := 1$

4243

$l := 2$

5623

$l := 3$

7410

$l := 4$

1521

$l := 5$

1560

$l := 6$

$l := 0$

80

```

l := 1
140
l := 2
186
l := 3
l := 0
l := 0
l := 0

```

(4)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(4*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],4*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],4*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],4*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],4*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],4*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(4*d)}$ 

Systemey4d := {}:

for k from 0 to (m-4)/3 do

Coeffs_y4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
k], y, 4*d)), [x,y])}:

Systemey4d := Systemey4d union Coeffs_y4d[4,m-4-3*k,k]:

od:

Systeme_y4d := Systemey4d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(4*d)}$ 

Variables_y4d := indets(Systeme_y4d):

```



```

## RESOLUTION  $y^{(4*d)}$ 

Solutions_y4d := solve(Systeme_y4d, Variables_y4d):
assign(Solutions_y4d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
116
l := 1
292
l := 2
418
l := 3
470
l := 4
438
l := 5
1742
l := 6
l := 0
48
l := 1
94
l := 2
122
l := 3
l := 0
l := 0
l := 0

```

(5)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(5*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*xiyj, i=0..dA[l]-j), j=0..min(dA

```

```

[l],5*d-1)): od:

for l from 0 to m - 1*3 do
BB[l] := add(add(B[l][i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],5*d-1)): od:

for l from 0 to m - 2*3 do
CC[l] := add(add(C[l][i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],5*d-1)): od:

for l from 0 to m - 3*3 do
DD[l] := add(add(D[l][i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],5*d-1)): od:

for l from 0 to m - 4*3 do
EE[l] := add(add(E[l][i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],5*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(5*d)

Systemey5d := {}:

for k from 0 to (m-5)/3 do

Coeffs_y5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,
k], y, 5*d)), [x,y])}:

Systemey5d := Systemey5d union Coeffs_y5d[5,m-5-3*k,k]:

od:

Systeme_y5d := Systemey5d:

## VARIABLES FOR DIVISIBILITY BY y^(5*d)

Variables_y5d := indets(Systeme_y5d):

## RESOLUTION y^(5*d)

Solutions_y5d := solve(Systeme_y5d, Variables_y5d):

assign(Solutions_y5d):

##

for l from 0 to m - 0*3 do nops(expand(AA[l])) od;
for l from 0 to m - 1*3 do nops(expand(BB[l])) od;
for l from 0 to m - 2*3 do nops(expand(CC[l])) od;
for l from 0 to m - 3*3 do nops(expand(DD[l])) od;
for l from 0 to m - 4*3 do nops(expand(EE[l])) od;

```

$l := 0$

```

l := 1
1
l := 2
1
l := 3
1
l := 4
1
l := 5
1
l := 6
l := 0
1
l := 1
1
l := 2
1
l := 3
l := 0
l := 0
l := 0

```

(6)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(6*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],6*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],6*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],6*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l],6*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l],6*d-1)): od:

```

```

## SYSTEM FOR DIVISIBILITY BY  $y^{(6*d)}$ 

Systemey6d := {}:

for k from 0 to (m-6)/3 do

Coeffs_y6d[6,m-6-3*k,k] := {coeffs(expand(mTaylor(Eq[6,m-6-3*k,
k], y,  $6^*d$ )), [x,y])}:

Systemey6d := Systemey6d union Coeffs_y6d[6,m-6-3*k,k]:

od:

Systeme_y6d := Systemey6d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(6*d)}$ 

Variables_y6d := indets(Systeme_y6d):

## RESOLUTION  $y^{(6*d)}$ 

Solutions_y6d := solve(Systeme_y6d, Variables_y6d):

assign(Solutions_y6d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
1
l := 2
1
l := 3
1
l := 4
1
l := 5
1
l := 6
l := 0
1
l := 1

```

```

1
l := 2
1
l := 3
l := 0
l := 0
l := 0

```

(7)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(7*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],7*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],7*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],7*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],7*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],7*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(7*d)}$ 

Systemey7d := {}:

for k from 0 to (m-7)/3 do

Coeffs_y7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,
k], y, 7*d)), [x,y])}:

Systemey7d := Systemey7d union Coeffs_y7d[7,m-7-3*k,k]:

od:

Systeme_y7d := Systemey7d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(7*d)}$ 

Variables_y7d := indets(Systeme_y7d):

## RESOLUTION  $y^{(7*d)}$ 

```

```
Solutions_y7d := solve(Systeme_y7d, Variables_y7d):
```

```
assign(Solutions_y7d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

```
l := 0
```

```
1
```

```
l := 1
```

```
1
```

```
l := 2
```

```
1
```

```
l := 3
```

```
1
```

```
l := 4
```

```
1
```

```
l := 5
```

```
1
```

```
l := 6
```

```
l := 0
```

```
1
```

```
l := 1
```

```
1
```

```
l := 2
```

```
1
```

```
l := 3
```

```
l := 0
```

```
l := 0
```

```
l := 0
```

(8)

```
>
```

```
## EXPAND ALL UNKNOWNNS UP TO y^(8*d)
```

```
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],8*d-1)): od:
```

```

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],8*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],8*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],8*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],8*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(8*d)

Systemey8d := {}:

for k from 0 to (m-8)/3 do

Coeffs_y8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,
k], y, 8*d)), [x,y])}:

Systemey8d := Systemey8d union Coeffs_y8d[8,m-8-3*k,k]:

od:

Systeme_y8d := Systemey8d:

## VARIABLES FOR DIVISIBILITY BY y^(8*d)

Variables_y8d := indets(Systeme_y8d):

## RESOLUTION y^(8*d)

Solutions_y8d := solve(Systeme_y8d, Variables_y8d):

assign(Solutions_y8d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
1
l := 2

```

```

1
l := 3
1
l := 4
1
l := 5
1
l := 6
l := 0
1
l := 1
1
l := 2
1
l := 3
l := 0
l := 0
l := 0

```

(9)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(9*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],9*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],9*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],9*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],9*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],9*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(9*d)}$ 

Systemey9d := {}:

```



```

for k from 0 to (m-9)/3 do
  Coeffs_y9d[9,m-9-3*k,k] := {coeffs(expand(mtaylor(Eq[9,m-9-3*k,
k], y, 9*d)), [x,y])}:
  Systemey9d := Systemey9d union Coeffs_y9d[9,m-9-3*k,k]:
od:
Systeme_y9d := Systemey9d:
## VARIABLES FOR DIVISIBILITY BY y^(9*d)
Variables_y9d := indets(Systeme_y9d):
## RESOLUTION y^(9*d)
Solutions_y9d := solve(Systeme_y9d, Variables_y9d):
assign(Solutions_y9d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
1
l := 2
1
l := 3
1
l := 4
1
l := 5
1
l := 6
l := 0
1
l := 1
1
l := 2

```

1
 $l := 3$
 $l := 0$
 $l := 0$
 $l := 0$

(10)

```
>
## EXPAND ALL UNKNOWNNS UP TO  $y^{(10*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],10*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],10*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],10*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l],10*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l],10*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(10*d)}$ 

Systemey10d := {}:

for k from 0 to (m-10)/3 do

Coeffs_y10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10-3*k,k], y, 10*d)), [x,y])}:

Systemey10d := Systemey10d union Coeffs_y10d[10,m-10-3*k,k]:

od:

Systeme_y10d := Systemey10d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(10*d)}$ 

Variables_y10d := indets(Systeme_y10d):

## RESOLUTION  $y^{(10*d)}$ 

Solutions_y10d := solve(Systeme_y10d, Variables_y10d):
```

```
assign(Solutions_y10d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

```
l := 0
```

```
1
```

```
l := 1
```

```
1
```

```
l := 2
```

```
1
```

```
l := 3
```

```
1
```

```
l := 4
```

```
1
```

```
l := 5
```

```
1
```

```
l := 6
```

```
l := 0
```

```
1
```

```
l := 1
```

```
1
```

```
l := 2
```

```
1
```

```
l := 3
```

```
l := 0
```

```
l := 0
```

```
l := 0
```

(11)

```
>
```

```
## EXPAND ALL UNKNOWNNS UP TO y^(11*d)
```

```
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],11*d-1)): od:
```

```
for l from 0 to m - 1*3 do
```

```

BB[l] := add(add(B[l][i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l], 11*d-1)): od:

for l from 0 to m - 2*3 do
CC[l] := add(add(C[l][i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l], 11*d-1)): od:

for l from 0 to m - 3*3 do
DD[l] := add(add(D[l][i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l], 11*d-1)): od:

for l from 0 to m - 4*3 do
EE[l] := add(add(E[l][i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l], 11*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(11*d)

Systemey11d := {}:

for k from 0 to (m-11)/3 do

Coeffs_y11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11-3*k,k], y, 11*d)), [x,y])}:

Systemey11d := Systemey11d union Coeffs_y11d[11,m-11-3*k,k]:

od:

Systeme_y11d := Systemey11d:

## VARIABLES FOR DIVISIBILITY BY y^(11*d)

Variables_y11d := indets(Systeme_y11d):

## RESOLUTION y^(11*d)

Solutions_y11d := solve(Systeme_y11d, Variables_y11d):

assign(Solutions_y11d):

##

for l from 0 to m - 0*3 do nops(expand(AA[l])) od;
for l from 0 to m - 1*3 do nops(expand(BB[l])) od;
for l from 0 to m - 2*3 do nops(expand(CC[l])) od;
for l from 0 to m - 3*3 do nops(expand(DD[l])) od;
for l from 0 to m - 4*3 do nops(expand(EE[l])) od;

l := 0
1
l := 1
1
l := 2
1

```

```

l := 3
1
l := 4
1
l := 5
1
l := 6
l := 0
1
l := 1
1
l := 2
1
l := 3
l := 0
l := 0
l := 0

```

(12)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(12*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],12*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],12*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],12*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l],12*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l],12*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(12*d)}$ 

Systemey12d := {}:

for k from 0 to (m-12)/3 do

```

```
Coeffs_y12d[12,m-12-3*k,k] := {coeffs(expand(mtaylor(Eq[12,m-12-3*k,k], y, 12*d)), [x,y])}:

```

```
Systemey12d := Systemey12d union Coeffs_y12d[12,m-12-3*k,k]:

```

```
od:

```

```
Systeme_y12d := Systemey12d:

```

```
## VARIABLES FOR DIVISIBILITY BY  $y^{(12*d)}$ 

```

```
Variables_y12d := indets(Systeme_y12d):

```

```
## RESOLUTION  $y^{(12*d)}$ 

```

```
Solutions_y12d := solve(Systeme_y12d, Variables_y12d):

```

```
assign(Solutions_y12d):

```

```
##

```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

```
l := 0

```

```
1

```

```
l := 1

```

```
1

```

```
l := 2

```

```
1

```

```
l := 3

```

```
1

```

```
l := 4

```

```
1

```

```
l := 5

```

```
1

```

```
l := 6

```

```
l := 0

```

```
1

```

```
l := 1

```

```
1

```

```
l := 2

```

```
1

```

```
l := 3

```

$l := 0$
 $l := 0$
 $l := 0$

(13)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(13*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],13*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],13*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],13*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l],13*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l],13*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(13*d)}$ 

Systemey13d := {}:

for k from 0 to (m-13)/3 do

Coeffs_y13d[13,m-13-3*k,k] := {coeffs(expand(mtaylor(Eq[13,m-13-3*k,k], y, 13*d)), [x,y])}:

Systemey13d := Systemey13d union Coeffs_y13d[13,m-13-3*k,k]:

od:

Systeme_y13d := Systemey13d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(13*d)}$ 

Variables_y13d := indets(Systeme_y13d):

## RESOLUTION  $y^{(13*d)}$ 

Solutions_y13d := solve(Systeme_y13d, Variables_y13d):

assign(Solutions_y13d):

##

```

```

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

$l := 0$

1

$l := 1$

1

$l := 2$

1

$l := 3$

1

$l := 4$

1

$l := 5$

1

$l := 6$

$l := 0$

1

$l := 1$

1

$l := 2$

1

$l := 3$

$l := 0$

$l := 0$

$l := 0$

(14)

> ## EXPAND ALL UNKNOWNNS UP TO $y^{(14*d)}$

```

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],14*d-1)): od:

```

```

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],14*d-1)): od:

```

```

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC

```



```

[l],14*d-1)): od:

for l from 0 to m - 3*3 do
DD[l] := add(add(D[l][i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],14*d-1)): od:

for l from 0 to m - 4*3 do
EE[l] := add(add(E[l][i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],14*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(14*d)

Systemey14d := {}:

for k from 0 to (m-14)/3 do

Coeffs_y14d[14,m-14-3*k,k] := {coeffs(expand(mtaylor(Eq[14,m-14
-3*k,k], y, 14*d)), [x,y])}:

Systemey14d := Systemey14d union Coeffs_y14d[14,m-14-3*k,k]:

od:

Systeme_y14d := Systemey14d:

## VARIABLES FOR DIVISIBILITY BY y^(14*d)

Variables_y14d := indets(Systeme_y14d):

## RESOLUTION y^(14*d)

Solutions_y14d := solve(Systeme_y14d, Variables_y14d):

assign(Solutions_y14d):

##

for l from 0 to m - 0*3 do nops(expand(AA[l])) od;
for l from 0 to m - 1*3 do nops(expand(BB[l])) od;
for l from 0 to m - 2*3 do nops(expand(CC[l])) od;
for l from 0 to m - 3*3 do nops(expand(DD[l])) od;
for l from 0 to m - 4*3 do nops(expand(EE[l])) od;

l := 0
1
l := 1
1
l := 2
1
l := 3
1
l := 4

```

```

1
l := 5
1
l := 6
l := 0
1
l := 1
1
l := 2
1
l := 3
l := 0
l := 0
l := 0

```

(15)

```

> ## HOLOMORPHICITY AT INFINITY
for l from 0 to m - 0*3 do
  HA||l := add(add(A||l[i,j]*X^i*Y^j*T^(dA[l]-i-j), i=0..dA[l]-j),
j=0..dA[l]):
od:
#
for l from 0 to m - 1*3 do
  HB||l := add(add(B||l[i,j]*X^i*Y^j*T^(dB[l]-i-j), i=0..dB[l]-j),
j=0..dB[l]):
od:
#
for l from 0 to m - 2*3 do
  HC||l := add(add(C||l[i,j]*X^i*Y^j*T^(dC[l]-i-j), i=0..dC[l]-j),
j=0..dC[l]):

```

```

od:
#
for l from 0 to m - 3*3 do
HD||l := add(add(D||l[i,j]*X^i*Y^j*T^(dD[l]-i-j), i=0..dD[l]-j),
j=0..dD[l]):
od:
#
for l from 0 to m - 4*3 do
HE||l := add(add(E||l[i,j]*X^i*Y^j*T^(dE[l]-i-j), i=0..dE[l]-j),
j=0..dE[l]):
od:
##
for j from 0 to m - 0*3 do
HK[j,0] := HA||j:
od:
#
for j from 0 to m - 1*3 do
HK[j,1] := HB||j:
od:
#
for j from 0 to m - 2*3 do
HK[j,2] := HC||j:
od:
#
for j from 0 to m - 3*3 do
HK[j,3] := HD||j:
od:
#
for j from 0 to m - 4*3 do
HK[j,4] := HE||j:

```

od:

> ##

```

printlevel := 2:
for q from 0 to m/3 do
for p from 0 to m-3*q do

HEq[p,q] := expand(mtaylor(add(add(add(
  (-1)^k * 2^(p-r) * (d+1)^(j+k-p-q) * binomial(j,r) * (k!)/((p-
r)!*q!*(k+r-p-q)!))
* X^(m-q-j-2*k) * T^(k-q) * RY^(2*k-2*q) * HK[j,k],
j=r..m-3*k),
r=max(0,p+q-k)..min(p,m-3*k)),
k=q..m/3)
, T, m-p-3*q)):
od: od;

```

```

q := 0
HEq := table([ ])
HEq0,0 := 0
HEq1,0 := 0
HEq2,0 := 0
HEq3,0 := 0
HEq4,0 := 0
HEq5,0 := 0
q := 1
HEq0,1 := 0
HEq1,1 := 0
HEq2,1 := 0
q := 2

```

(16)

```

> HSysteme := {seq(seq(coeffs(HEq[p,q],[T,X,Y]),p=0..m-3*q),q=0..
m/3)} minus {0}:

HSolutions := solve(HSysteme):

assign(HSolutions):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
1
l := 2
1
l := 3
1
l := 4
1
l := 5
1
l := 6
1
l := 0
1
l := 1
1
l := 2
1
l := 3
1
l := 0
1
l := 0
1
l := 0

```

```

> Jet_xR := factor(
+ (1/Ry^(m-0*2)) * x^(-twist) * add(AA||j * x1^(m-0*3-j) *
(R1/RR)^j * Delta_xR^0, j=0..m-0*3)
+ (1/Ry^(m-1*2)) * x^(-twist) * add(BB||j * x1^(m-1*3-j) *
(R1/RR)^j * Delta_xR^1, j=0..m-1*3)
+ (1/Ry^(m-2*2)) * x^(-twist) * add(CC||j * x1^(m-2*3-j) *
(R1/RR)^j * Delta_xR^2, j=0..m-2*3)
+ (1/Ry^(m-3*2)) * x^(-twist) * add(DD||j * x1^(m-3*3-j) *
(R1/RR)^j * Delta_xR^3, j=0..m-3*3)

```

$$\left[\begin{array}{l} + (1/Ry^{(m-4*2)}) * x^{(-twist)} * add(EE||j * x1^{(m-4*3-j)} * \\ (R1/RR)^j * Delta_x R^4, j=0..m-4*3)); \\ \\ Jet_xR := 0 \end{array} \right. \quad (18)$$

$$\left[\begin{array}{l} > Vars := indets(Jet_xR) minus \{x, x1, x2, y, R1, R2\}; \\ NbVars := nops(Vars); \\ \\ Vars := ; \\ NbVars := 0 \end{array} \right. \quad (19)$$

$$\left[\begin{array}{l} > Ens_Vars := \{\}: \\ \text{for } i \text{ from } 1 \text{ to } nops(Vars) \text{ do} \\ Ens_Vars := Ens_Vars \text{ union } \{op(i,Vars)=0\}: \\ \text{od}; \\ EnsVars := Ens_Vars; \\ \\ EnsVars := : \end{array} \right. \quad (20)$$

$$\left[\begin{array}{l} > \text{for } i \text{ from } 1 \text{ to } nops(Vars) \text{ do} \\ JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jet_xR))): \\ \text{od}; \\ \\ i := 1 \end{array} \right. \quad (21)$$